

Comment on “Request for Information: Accelerating the Adoption and Use of Artificial Intelligence as Part of Clinical Care.”

A Deterministic Semantic Validation Layer to Support Safe AI and Longitudinal Interoperability

Executive Summary

The U.S. healthcare ecosystem has made substantial progress in enabling electronic data access and exchange, particularly through increasing adoption of FHIR-based APIs. These advances have improved patient-centered access, enabled third-party innovation, and laid the groundwork for AI-enabled clinical tools.

As AI and machine learning systems increasingly operate downstream of interoperable data flows, a new technical challenge emerges—ensuring that clinical data and quality measurement inputs retain consistent semantic meaning as they move across systems, organizations, and time. While data transport is increasingly standardized, semantic interpretation is not. Semantic interoperability gaps—differences in units, measurement context, specimen, method, or transformation rules—can affect AI reasoning, longitudinal analysis, and the interpretation of trends across patient populations.

This comment proposes exploring a deterministic, hybrid human–AI semantic validation layer: a complementary, auditable framework designed to stabilize clinical meaning for AI-enabled analysis, longitudinal monitoring, and quality learning, without constraining innovation, and to extend the value of existing FHIR-based interoperability.

1. The Emerging Challenge: Semantic Drift and Context Loss in an Interoperable, AI-Enabled Ecosystem

FHIR-based APIs have succeeded in enabling data movement. However, interoperability at the transport layer does not guarantee semantic equivalence at the interpretation layer.

As data moves across systems, system-specific transformations may occur upstream—including unit conversions, aggregation, normalization, or omission of contextual attributes. These

transformations may be valid within a local system, given local configuration and reference context, but may not be visible, traceable, or interpretable downstream.

These issues are often manageable within:

- a single clinical endpoint
- a limited-radius application ecosystem
- or a single EHR context

However, they become increasingly problematic when:

- patients move between health systems
- data is compared longitudinally across years
- multiple AI models consume the same data
- or research and monitoring span heterogeneous sources

2. When Data Cannot Be Compared at All

In some cases, two clinical values cannot be meaningfully compared under the rules of measurement science without explicit transformation rules.

For example, a laboratory result for serum iron produced under one assay method and specimen matrix cannot be meaningfully compared to a result produced under a different method or matrix unless the underlying measurand and validated harmonization logic are known; unit conversion alone is insufficient when the measurement procedure differs. In such cases, longitudinal comparison is not merely imprecise—it is invalid.

AI systems, particularly large language models, will often attempt to infer over whatever data is provided. Without explicit semantic context and deterministic rules, these systems may infer continuity where none exists.

This behavior is often described as hallucination; however, in many cases the root cause is missing semantic ground truth rather than model failure.

When clinical data lacks explicit semantic definition—such as standardized units, measurement conditions, or clearly defined reference meaning—AI systems must rely on probabilistic inference to interpret gaps or inconsistencies. While probabilistic reasoning is not inherently unsafe, it limits precision and reproducibility, particularly as data moves across systems or is analyzed over time. By contrast, deterministic semantic validation ensures that values, transformations, and interpretation boundaries are explicit and auditable, reducing reliance on inference and enabling

AI to operate on known, well-defined inputs rather than assumptions. This distinction becomes increasingly important as healthcare moves from isolated clinical snapshots toward longitudinal monitoring, where small semantic differences compound over time and can otherwise prevent meaningful comparison, trend analysis, or early pattern detection.

3. Hybrid Deterministic Semantic Validation Layer

A hybrid deterministic semantic validation layer introduces explicit, auditable rules that define how clinical data is interpreted for downstream use—particularly for AI-enabled analysis and longitudinal monitoring.

Key characteristics include:

- deterministic processing for unit normalization, reference alignment, and semantic validation
- explicit provenance and audit logs capturing transformations, assumptions, and unresolved gaps
- a hybrid design in which AI assists but does not replace human oversight

Clinical reference catalogs can be derived or synthesized with assistance from large language models using clearly defined source constraints, followed by human expert review, approval, and versioned publication as shared benchmarks.

This approach does not constrain AI; it supports safer and more reproducible AI reasoning.

4. Why This Enables AI Rather Than Restricting It

AI and machine learning systems are highly flexible, but they will attempt to infer meaning when semantic gaps are not explicitly defined.

A deterministic semantic layer:

- prevents implicit inference when data is incomplete or non-comparable
- enables AI systems to explicitly represent and communicate uncertainty
- allows machine learning models to operate on semantically stable, well-defined inputs

Large language models can synthesize clinical context, documentation, and explanations. Machine learning models can validate hypotheses, detect trends, and surface emerging patterns. Both benefit from a shared, explicit semantic foundation that reduces ambiguity and improves reliability.

5. Concrete Example: Semantic Equivalence in Longitudinal Laboratory Monitoring

Consider a patient whose triglyceride values are reported over time by different laboratories:

- one using mg/dL
- another using mmol/L
- with differing reference ranges and assay context

Without explicit normalization rules and provenance:

- longitudinal trends may be misinterpreted
- clinical thresholds may be applied inconsistently
- and AI-generated insights may appear contradictory

With deterministic semantic validation:

- values are converted using validated rules appropriate to the underlying measurand
- reference ranges and interpretation context are applied consistently
- and AI systems are explicitly aware of what is known, transformed, or missing

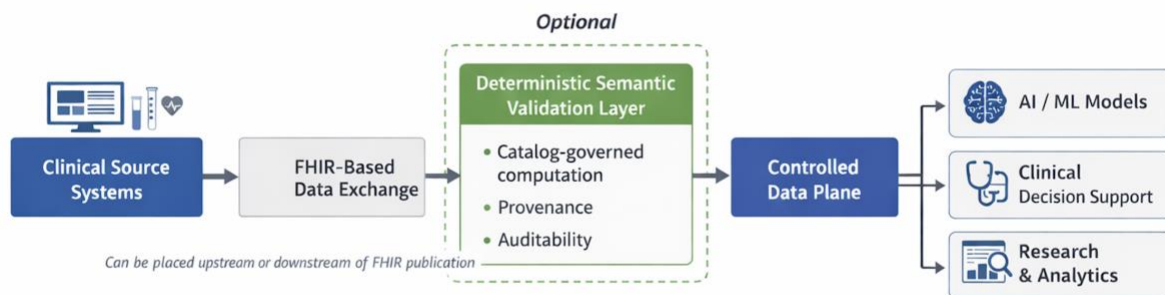
In other laboratory contexts—such as measurements based on enzyme activity versus mass (e.g., CK-MB), biological potency (e.g., insulin, FSH), or immunoassay-based versus mass spectrometry-based results (e.g., testosterone)—differences in reported units reflect fundamentally different measurands or measurement bases. In these cases, mathematical conversion is scientifically invalid, and values should not be compared.

To preserve data integrity, non-comparable results should be represented using distinct LOINC codes or equivalent semantic identifiers, ensuring that automated systems do not normalize or compare incompatible measurement bases.

FHIR provides flexible mechanisms—such as the Observation and Provenance resources—to capture structured measurements, attribution, and contextual metadata, while leaving transformation logic and semantic interpretation to implementers. As healthcare data exchange increasingly supports longitudinal monitoring, cross-system reuse, and AI-enabled analysis, a governed, hybrid semantic validation layer upstream can complement these capabilities by stabilizing semantic interpretation and validated transformation logic at the point of reuse.

In practice, this enables data pipelines to apply method-specific semantic differentiation before longitudinal analysis occurs, improving precision and consistency over time and making provenance and audit information more interpretable as data is reused at scale.

6. Placement Within the Interoperability Stack



Illustrative workflow showing where deterministic semantic validation can complement FHIR-based exchange.

This semantic validation layer does not require a single architectural placement. In general, placement of the deterministic semantic validation layer closer to data origination enables more precise semantic validation and higher downstream impact, while later placement can still provide meaningful stabilization for longitudinal reuse.

It can be deployed, independently or in combination:

- within a clinical source system prior to external data exposure
- immediately before FHIR-based API publication
- or downstream of FHIR consumption by applications or AI systems

This flexibility allows stakeholders to evaluate where deterministic semantic stabilization is most effective as interoperability and data reuse scale, without constraining architectural choice or prescribing a single deployment model.

7. Applicability Beyond Laboratory Data

While laboratory data and medications provide the most explicit semantic starting points, the same principles apply to:

- clinical observations
- device-generated data
- and longitudinal clinical assessments

Each domain benefits from explicit semantic context as data is reused across systems and over time, particularly for AI-enabled decision support, longitudinal monitoring, and research. In these domains, determinism is achieved through explicit semantic constraints and comparability rules rather than purely mathematical normalization.

8. Relationship to Current Progress

Significant progress has already been achieved:

- health data is more portable than ever
- patient access initiatives continue to expand
- clinicians are increasingly supported by interoperable tools
- and AI and machine learning systems are already helping patients better understand health information, navigate care decisions, and engage with the healthcare system, while also supporting analytical and clinical workflows

The remaining challenge is not ambition, but orchestration—aligning data movement, semantic clarity, and AI reasoning into a coherent, auditable workflow that supports longitudinal use and reuse over time.

9. Collaborative Evaluation Opportunity

Participation in such an approach could be complementary to existing FHIR-based interoperability. Coordinated, multi-stakeholder evaluation efforts could help assess where semantic validation is most effective across the data exchange pipeline, how deterministic layers evolve over time, and how AI system reliability and longitudinal comparability improve when explicit semantic context is preserved. HHS could consider providing guidance or serving as a coordinating convener for such efforts as a means of enabling shared learning without disrupting existing implementations.

In addition, HHS could play a constructive role in convening clinical and standards bodies to help establish governance for the clinical reference inputs used in deterministic semantic layers. This could include supporting clinician-led review and sign-off on source materials, reference ranges, and transformation rules, as well as encouraging the synthesis of shared benchmark references. Once established, such clinically governed reference layers could be reused across the ecosystem, enabling consistency, transparency, and trust while preserving innovation and implementation flexibility.

Conclusion

As the healthcare ecosystem continues to evolve from isolated data exchange toward longitudinal, AI-assisted monitoring and discovery, semantic clarity becomes foundational infrastructure.

A hybrid deterministic semantic validation layer could provide a pragmatic, incremental approach to supporting safer AI reasoning, more reliable longitudinal analysis, and continued innovation—while complementing existing interoperability standards and preserving architectural flexibility.

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